

INTERFACE REBUILD • 2026

# One estate, four jobs, one interface that tells you what needs you.

A full visual and interaction rebuild of CommuniKey, the safety, access and community platform for South African residential estates. Every screen, for all four roles, rebuilt on a single design system without changing a line of what the product does.

DESIGN AND BUILD

**Leseli Morakile**

SCREENS REBUILT

**16 pages • 4 roles • 1 system**

## Four different jobs, sharing one record of what happened

CommuniKey runs the security and community side of a residential estate. A resident issues a pass for a guest. The guard at the boom checks that pass against the register. When something goes wrong, a safety volunteer picks it up and works it to a close. Estate management verifies who actually lives there and reads the numbers at the end of the month.

### Residents

Issue visitor passes, report incidents, raise an SOS, RSVP to estate events.

### Safety volunteers

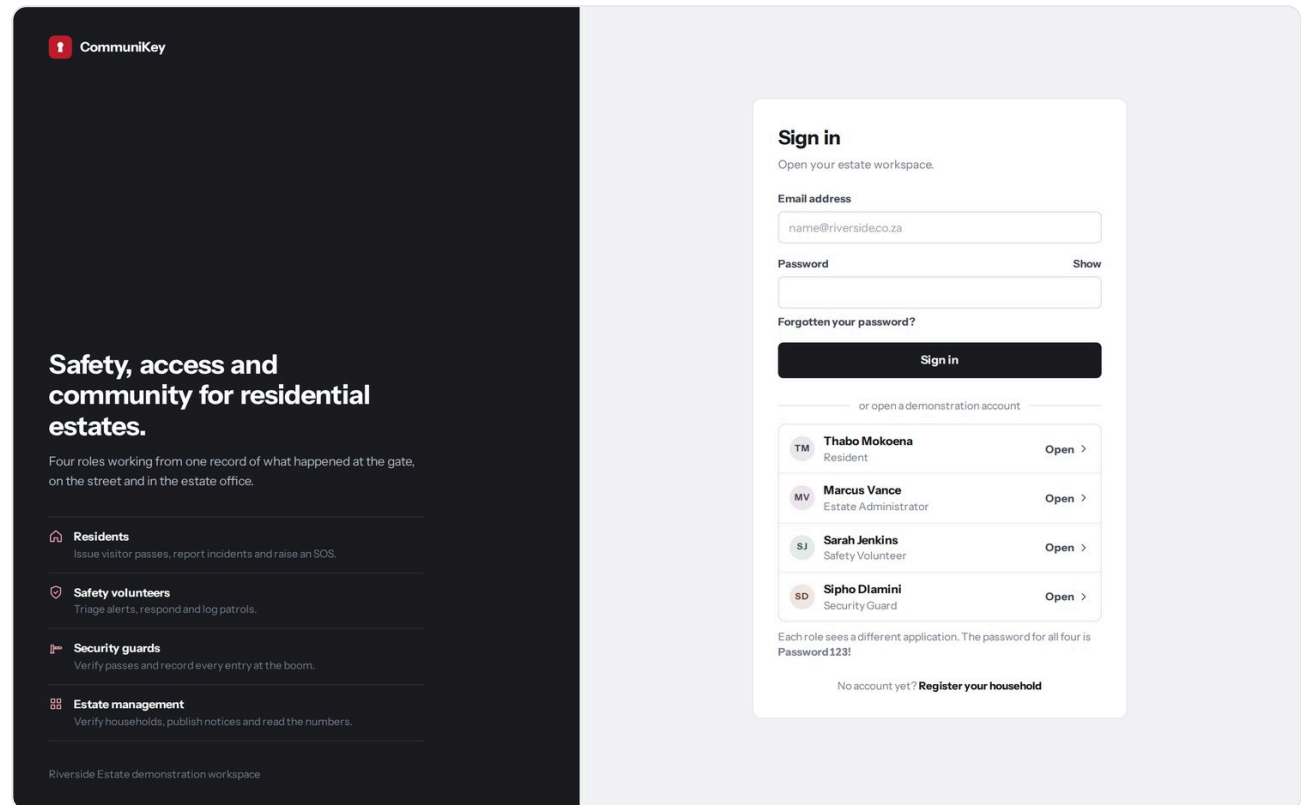
Triage incoming reports, move an SOS through its response stages, log patrols.

### Security guards

Verify a PIN or scan a code at the boom, authorise entry, keep the gate ledger.

### Estate management

Verify households against proof of residence, publish notices, read the reporting.



Sign in. The four roles are stated on the panel and each one can be opened directly, because a reviewer should not have to memorise four addresses to discover that the product has four different applications inside it.

## The interface did not agree with itself

The product worked. Reading it was the problem. Each screen had been built on its own terms, so the same idea looked different depending on where you found it, and colour had stopped carrying any meaning at all.

### 01

#### Colour meant nothing

Thirteen status colours were in use across the screens, several of them for states that meant roughly the same thing. When everything is coloured, nothing reads as urgent, which is a real cost in a product whose whole job is to surface the one thing that needs a person now.

### 02

#### No shared furniture

Tables, cards, buttons and empty states were re-invented page by page. A resident moving from the incident list to the directory was learning the same layout twice, in two different shapes.

### 03

#### The urgent thing floated

The SOS control was a floating button parked over the content, the pattern normally reserved for composing a message. Confirmations pushed the page down as they appeared, so the row you were about to click moved out from under the cursor.

#### What the rebuild had to protect

Every flow, permission rule and figure already in the product. The masking that hides a neighbour's phone number from other residents, the role based routing that keeps a guard out of the reporting, the seeded 84 days of incident and gate history that all the analytics are computed from. None of it was to be touched.

#### What it had to change

The visual language, in full, to the dense administrative style of the supplied reference: a dark grouped sidebar, a light working canvas, section headers that carry their own figures, and restrained colour used only where it signals something real.

## Four rules, applied to every screen without exception

---

01

### One colour, one word, one meaning

The interface is charcoal, white and grey. Seven states carry the colour, each with a single word that holds on every screen and in every chart, so a person never has to work out whether two amber labels mean the same thing.

02

### Answer "what needs me" above the fold

Every role's first screen opens with the things waiting on that person, with the counts in the header bar. Reporting and history sit below it. The administrator sees the verification queue before the charts; the volunteer sees the live SOS before the patrol roster.

03

### One piece of furniture per job

One section bar, one card, one table, one status pill, one empty state, one dialog. Tables are declared as columns and rows through a single component, so a column cannot drift out of line with the same column on another screen.

04

### Nothing moves under the hand

Confirmations appear as a message at the foot of the screen rather than pushing content down. Dialogs place the cursor in the first field and leave it there. Status changes settle in place. The layout stays where the person left it.

---

38 → 7

status words, one per colour, product wide

16

pages rebuilt on the shared system

41

end to end flows tested across the four roles

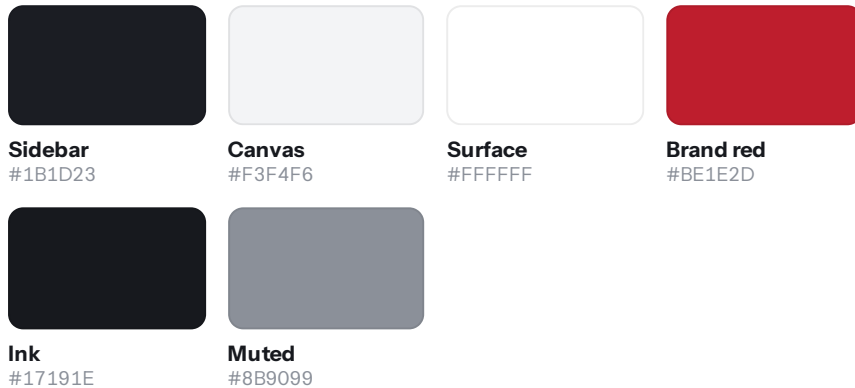
0

features removed or changed in behaviour

## A restrained palette, one typeface, seven meanings

### Palette

Charcoal holds the navigation, white holds the work, grey separates them. Red is the identity and the one data signal.



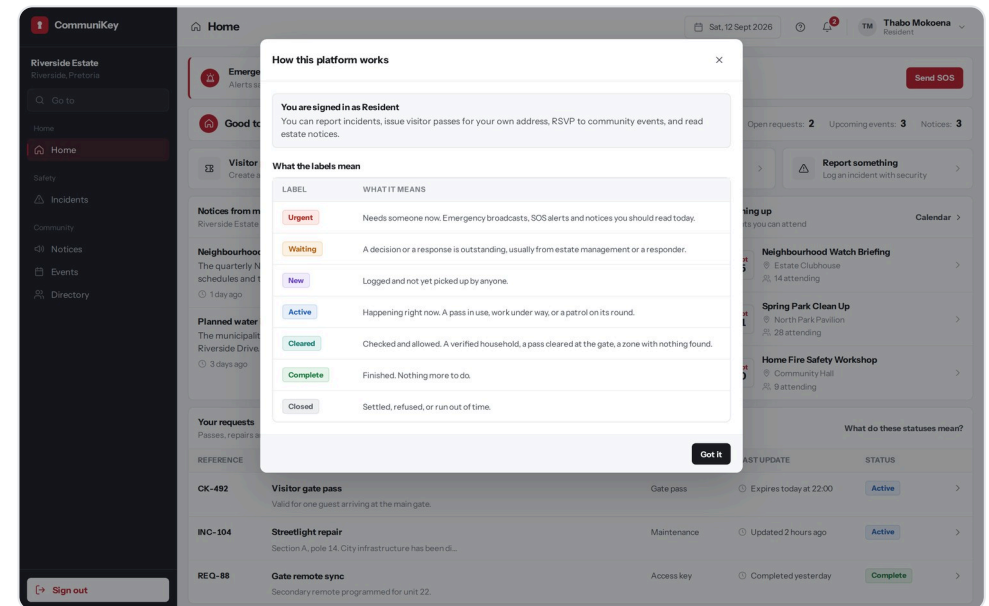
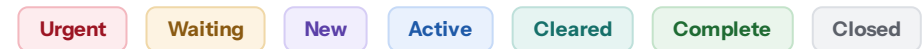
### Typeface

Instrument Sans, served by the application itself rather than fetched from a font service, so a guardhouse terminal with no outside connection renders exactly the same interface. Figures are set with tabular numerals so columns of counts line up.

|          |                                               |
|----------|-----------------------------------------------|
| 30 / 700 | <b>Estate overview</b>                        |
| 15 / 600 | <b>Waiting on you</b>                         |
| 13 / 600 | <b>Verification queue</b>                     |
| 12 / 400 | New households waiting for proof of residence |
| 11 / 600 | <b>REPORTED BY</b>                            |

### The status system

Thirty eight status words in thirteen colours became seven states. Each state has one word and one colour, everywhere, so a colour never carries two meanings and a person learns the vocabulary once. Records keep their own values; one component turns them into the word a person reads, so no screen can invent an eighth.



The legend is not buried in a specification. It is in the product, one click from the top bar on every screen: seven words, seven colours, and what each one means.

## Resident home

The four things a resident actually does are stated as plain actions across the top: issue a pass, check a code, call the gate, report something. Under them sit the notices, what is coming up, and the status of anything they have already asked for.

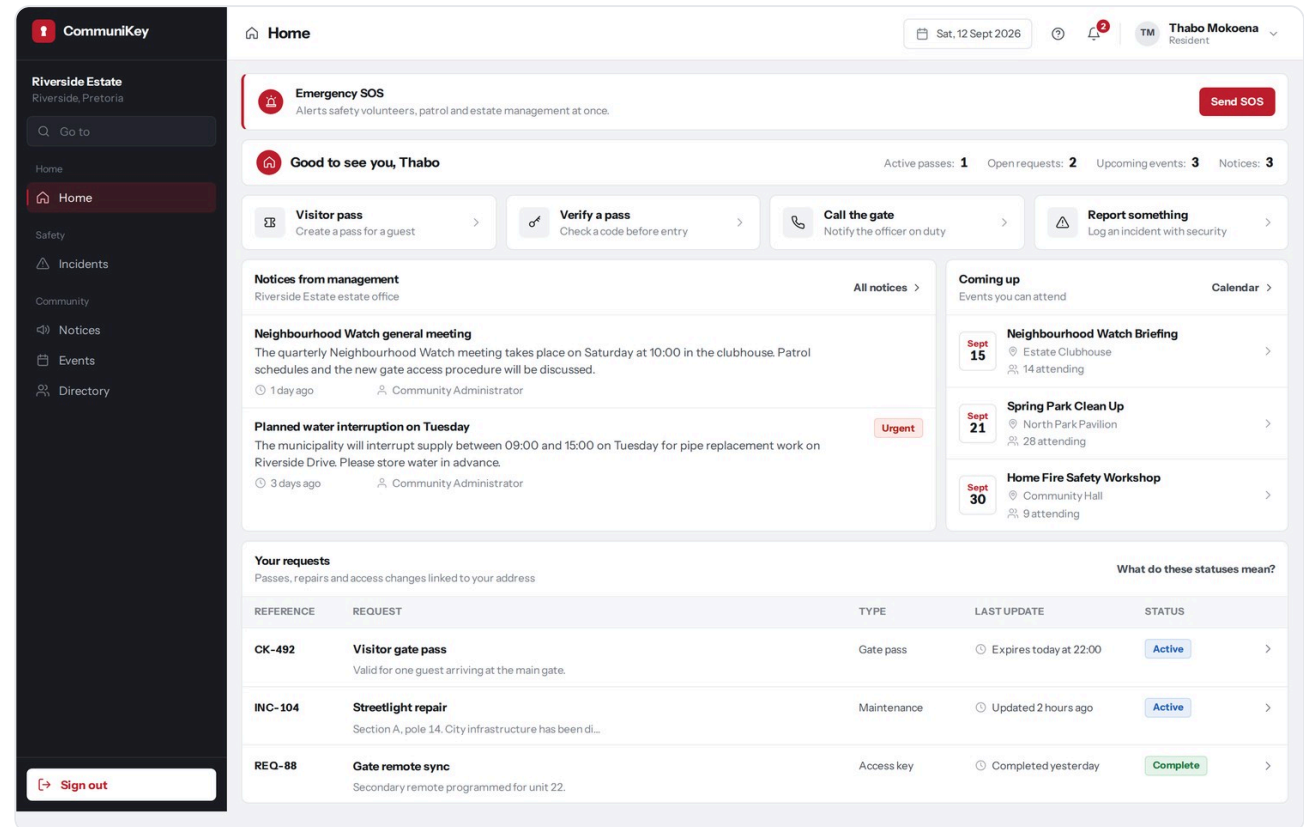
### The SOS is a strip, not a floating button

It sits at the top of the page, full width, with its consequence written next to it: this alerts safety volunteers, patrol and estate management at once.

Sending it takes a confirmation and then a countdown that can still be stopped, because the cost of a false alarm falls on real people.

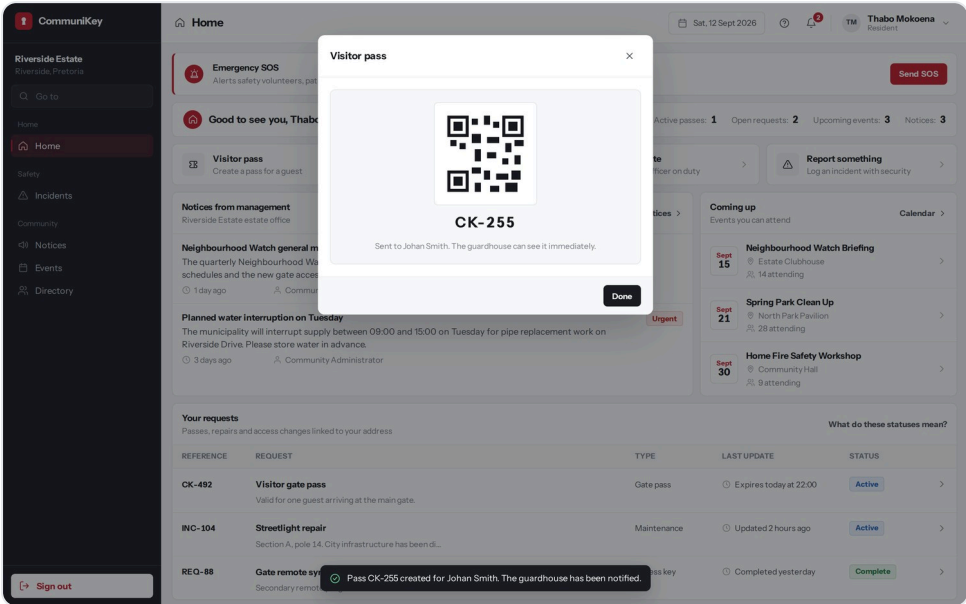
### Every request carries its own reference

CK-492, INC-104, REQ-88. A resident phoning the guardhouse has something to read out, and the same reference appears on the guard's ledger.

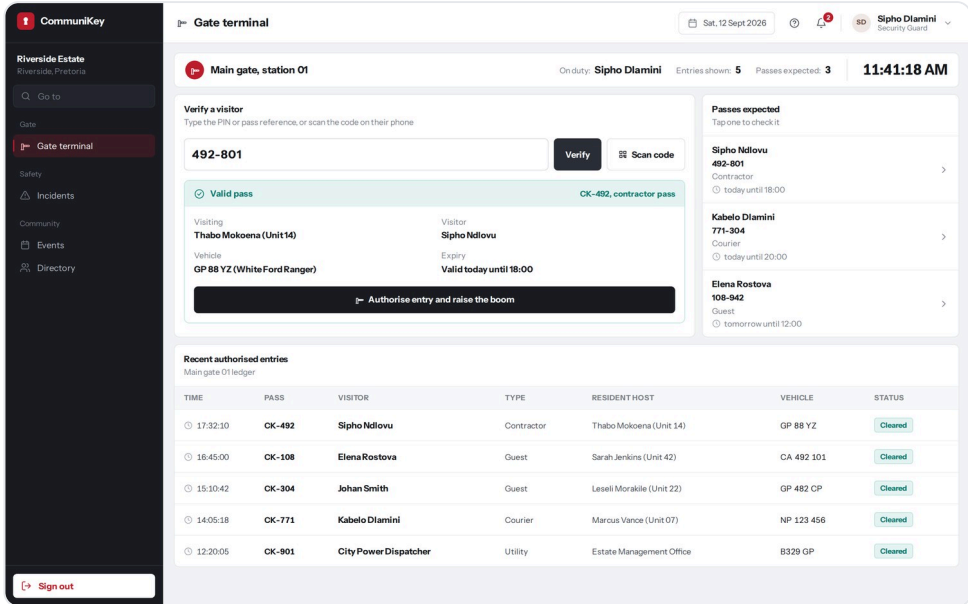


Resident home at 1440 wide.

# Issuing a pass, and the moment it is used



The resident gets a code and a scannable pass, and is told plainly that the guardhouse can already see it.



The same pass at the boom. The verdict panel is the largest element on the guard's screen.

## The guard's screen is a single question

Does this person come in or not. The PIN field is oversized and focused the moment the page loads, so an officer can type without touching the mouse. The answer comes back as a panel, not a line of text: who the pass belongs to, which resident issued it, and until when. Authorising writes the entry to the ledger below.

## A refusal has to be as clear as an approval

An expired or unknown code returns a red panel that says access denied and why. There is no ambiguous middle state for an officer to interpret under pressure at a gate with a car waiting behind the boom.

## Estate overview

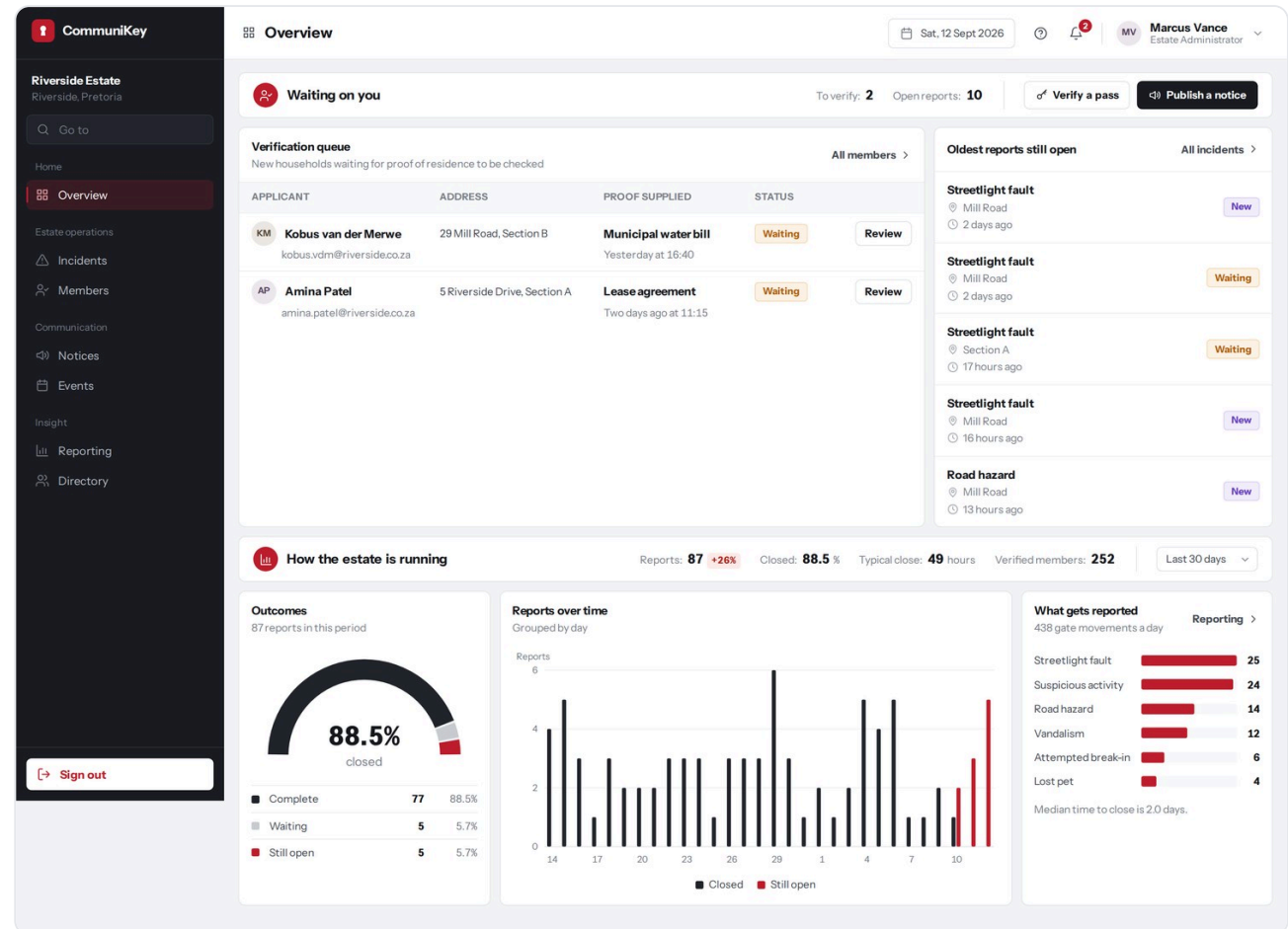
Management opens on the two things only they can clear: households waiting for their proof of residence to be checked, and the reports that have been open longest. The reporting sits underneath, behind its own section bar.

### Figures live in the header bar

Each section states its own counts inline, the pattern taken from the reference. The header is the summary, so the page does not need a row of large number tiles competing with the work below it.

### Red only means still open

In the outcome gauge and the daily bars, charcoal is closed work and red is what remains. The eye goes to the last few days of the chart, which is exactly where an administrator's attention belongs.



Estate overview, administrator.



# Reporting

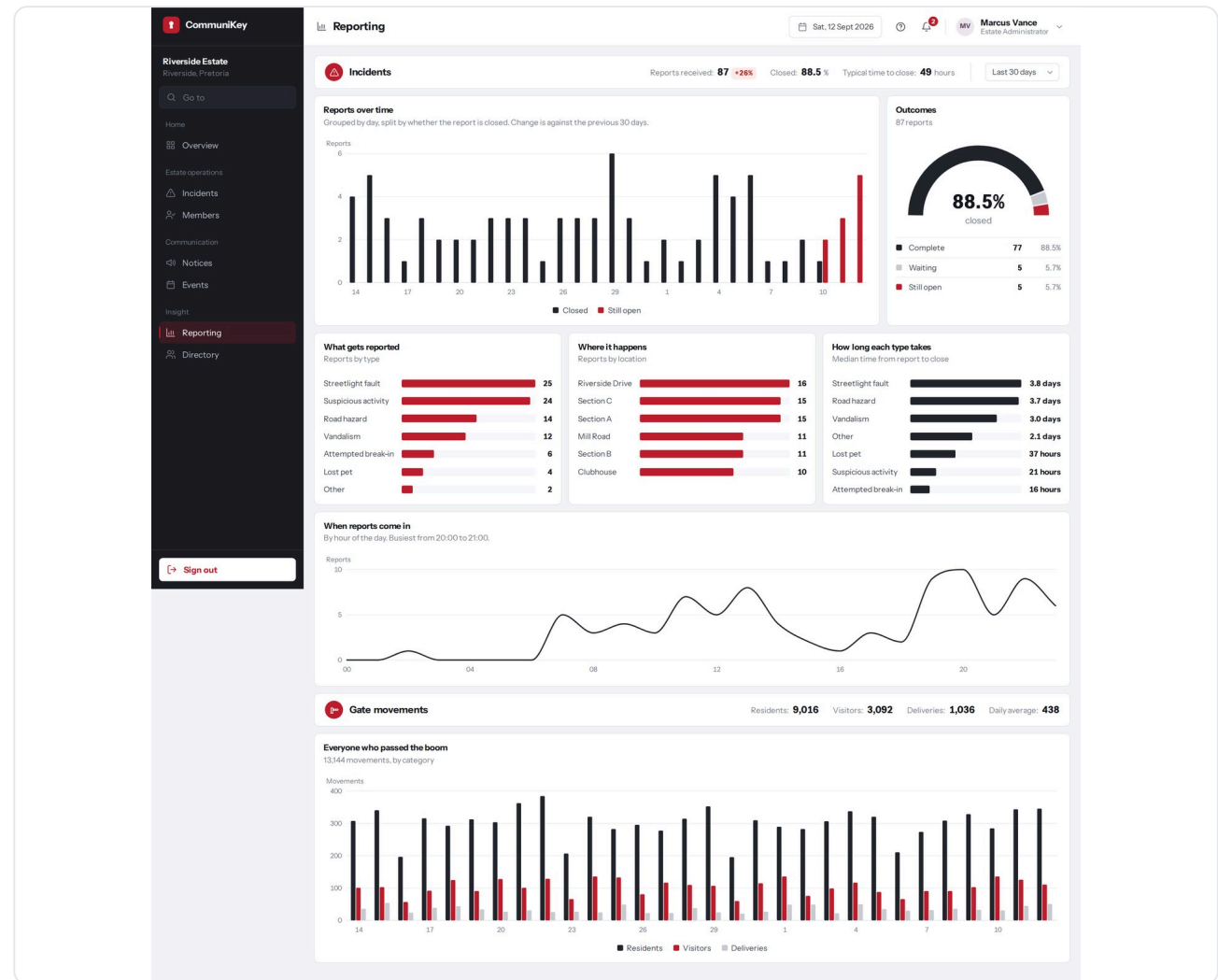
Every figure here is computed from the seeded incident and gate history by the existing analytics layer. The rebuild changed how the numbers are drawn, never what they are.

## Charts drawn at their real size

The charts measure the space they are given and draw at that width, rather than being scaled up from a fixed drawing. Labels stay at the size they were designed at instead of stretching, and an axis of report counts never shows a tick like 3.8 of an incident.

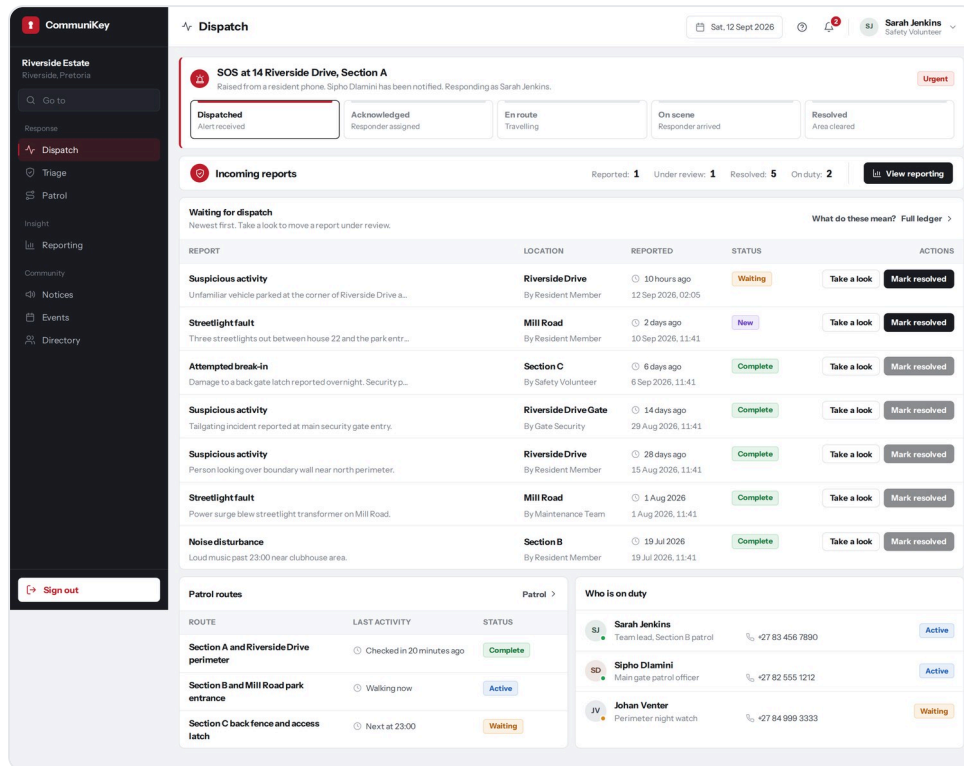
## Ranked bars instead of pie charts

What gets reported, where it happens, and how long each type takes to close are all ordered lists with a bar and a figure. Comparing lengths is easier than comparing angles, and the labels stay readable.



Reporting, full page.

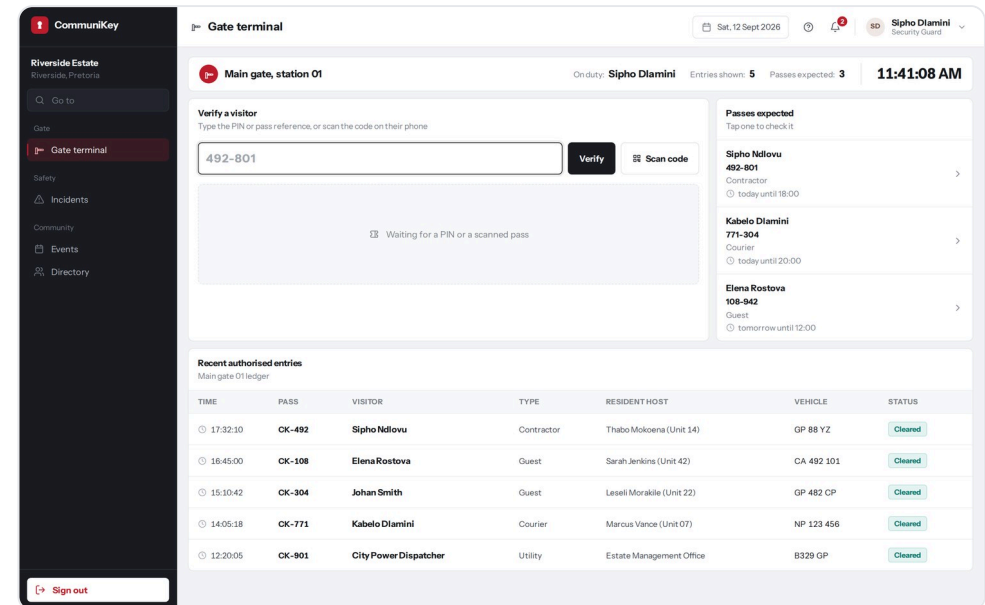
# Dispatch and the gate



Volunteer dispatch. The live SOS and its response stages sit above the queue.

## A response has stages, so the interface has stages

Dispatched, acknowledged, en route, on scene, resolved. The volunteer taps the stage they have reached and the resident who raised the alarm sees it move. Previously this was a status word in a dropdown, which told nobody how far along anything was.

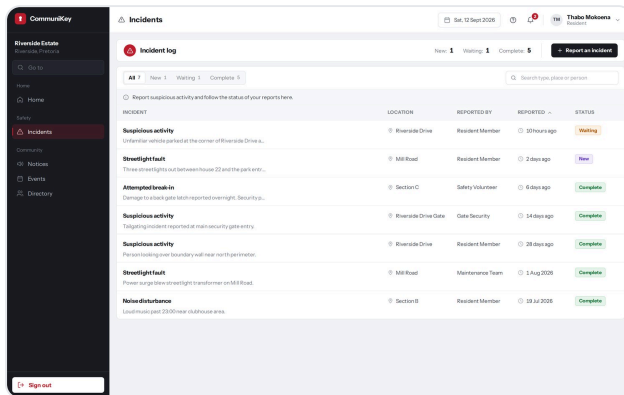


Gate terminal. Four sections only, and a running clock on the ledger.

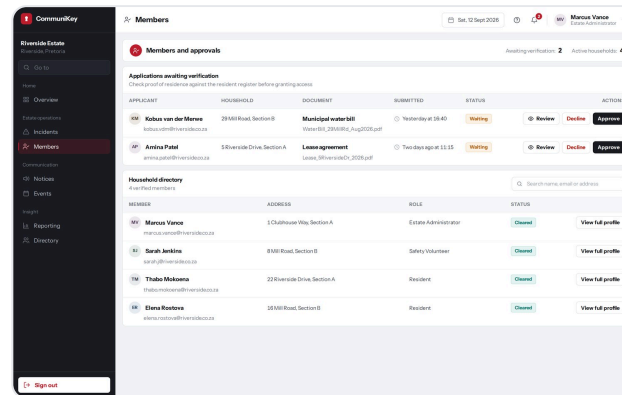
## Each role only sees their own sections

A guard's sidebar holds four items and no reporting. Typing an administrator's address into the bar returns them to their own screen rather than to a dead end. The permission rules are the original ones; the navigation now reflects them honestly.

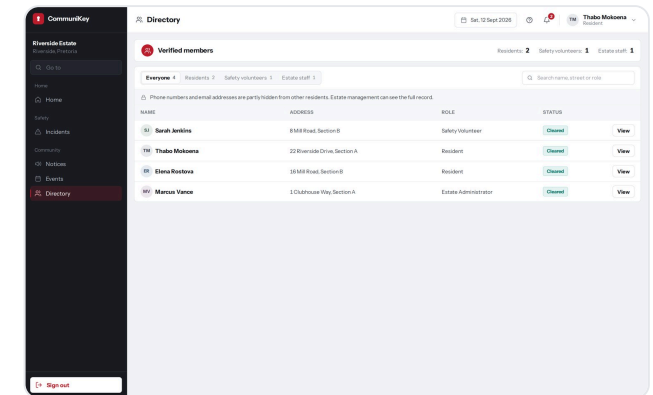
# Lists, queues and the small screen



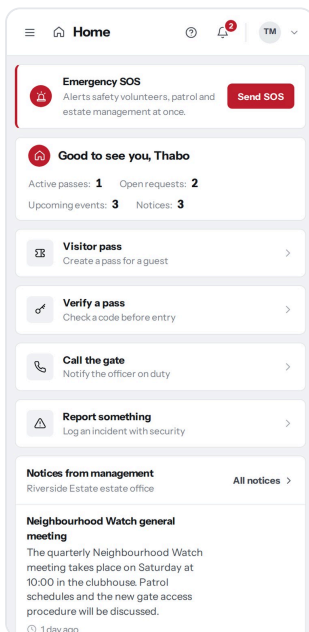
Incidents. Tabs carry their own counts, so a person sees the size of a view before choosing it.



Members. The approvals queue and the verified directory on one screen, in that order.



Directory. A line states plainly that contact details are partly hidden and who can see the full record.



## The same system at 390 wide

The sidebar becomes a drawer that closes itself once a section is chosen. A table stops being a table below a threshold set by how many columns it has: each row becomes a card, with the first column as the heading and the rest as labelled facts whose labels share a column. Nothing scrolls sideways, and the type does not shrink below the size it was designed at. Checked at every width from 360 to 2560.

## Empty is a state, not an accident

Every list has a written empty state with a way out of it. A search with no matches offers to clear itself. A cleared approvals queue says so, rather than showing an empty table with headers and nothing underneath.

## Eight changes that were about behaviour, not appearance

### The emergency control

#### BEFORE

A floating button sitting over the page content, the pattern normally used for composing something new.

#### AFTER

A strip at the top of the resident home that states who it alerts, with a countdown that can still be stopped.

### Typing in a dialog

#### BEFORE

Focus was pulled back to the first field on every keystroke, so a name could not be typed in one go.

#### AFTER

The cursor is placed once when the dialog opens and left alone. Escape closes it.

### Getting back to dispatch

#### BEFORE

The volunteer dispatch screen had no link of its own. It was reachable only by clicking the logo.

#### AFTER

It is the first item in that role's sidebar, and the sidebar lists only sections that role can open.

### Columns

#### BEFORE

Each table was written out by hand, so the same column was a different width on each screen and cells floated against the middle of a wrapped row.

#### AFTER

One table component. A column declares its width and its alignment once, and every cell on a row sits on the same line.

### Confirmations

#### BEFORE

Messages appeared inside the layout and pushed the rows below them down the page.

#### AFTER

A message at the foot of the screen. Nothing moves, so the row under the cursor stays under the cursor.

### Status words

#### BEFORE

Thirty eight status words across thirteen colours, so one colour carried four different labels and a person had to learn them screen by screen.

#### AFTER

Seven states, each with one word and one colour, turned out by a single component so no screen can invent an eighth.

### The approvals queue

#### BEFORE

The overview and the members screen each kept their own copy, so approving in one left it sitting in the other.

#### AFTER

One shared queue. Clearing an application anywhere clears it everywhere, which is what an administrator expects.

### Labels in a sentence

#### BEFORE

A priority label sat inside the title, interrupting the sentence and leaving the titles below it starting at different places.

#### AFTER

The label sits on its own line above the title, so every title in the list begins at the same left edge.

## What shipped

- Sixteen pages rebuilt on one design system, covering all four roles and every dialog, empty state and interaction state.
- A component layer that did not exist before: section bars, cards, the table, status pills, dialogs, toasts, tabs, search fields, empty states and charts, each built once.
- Every cell's first line is a box of one fixed height, so words, pills, timestamps and avatars sit on the same line no matter how a row wraps.
- Facts in a list claim a column of their own, so the second and third fact start at the same place on every row instead of following the one before it.
- Charts drawn in plain SVG with no charting dependency added, reading the same analytics the product already computed.
- The typeface served by the application itself, so the interface renders identically on a terminal with no outside connection.
- Forty one end to end flows tested across the four roles, from sign in through to the gate ledger, with no console errors.
- Six unit tests covering sign in, the role based shell and the permission redirect.
- No feature removed, no permission relaxed, no figure altered. The masking, the role routing and the seeded history are the originals.
- Three known bugs fixed on the way through, listed on the previous page.

16

pages

4

roles

41

flows tested

7

status tones

1

design system

